

DS 6021: Intro to Predictive Modeling

Lab 5: Feature Selection, Cross-Validation, and Model Complexity

October 9, 2026

Due: October 15, 2026 11:59pm

Materials Required

- Auto.csv – Gas mileage (mpg), horsepower, and other information for cars from model years 1970–1982, from An Introduction to Statistical Learning (ISL). This is a version of the UCI Auto MPG dataset.
 - Note: There are also worked examples for this dataset in ISL.
- bike-sharing-daily.csv – Daily bike rental counts, Washington DC, 2011–2012
- In-class notebooks from 9/30 (feat_selection_pt3_cross_validation.ipynb) and 10/7 (poly_regression_and_splines.ipynb)
- Week 5 + 6 slides (Feature Selection + Model Complexity) and Week 7 slides (Polynomial and Spline Regression)

Activity Overview

For the last few weeks, we have been learning about several approaches to answer the following question: how do we build a model that is flexible enough to capture the patterns in our training data, but that still works on data it has never seen? To do this, we have covered the following techniques and concepts:

- Feature Selection: Forward, Backward, and Recursive Feature Elimination
- Holdout sets: Train/test splits and cross-validation
- Moving beyond linear models using polynomial and splines
- The bias-variance tradeoff as a conceptual framework that helps pull all of this together.

In this lab we will explore all of these concepts in a bit more detail.

In Part 1, we will look at feature selection techniques using the Auto dataset. Our model will be to predict miles per gallon (mpg) based on other variables in the dataset.

In Part 2, we will once again use the Auto dataset, but this time use cross-validation to compare different models.

In Part 3, we will continue learning about polynomial and spline regression. This part picks up where we left off in class on 10/7.

Finally, in Part 4, we will do a bootstrapping exercise to further explore the bias-variance tradeoff.

Part 1: Feature Selection (10 pts)

In class on 9/30, we compared forward selection, backward selection, RFE, and RFECV for predicting the daily count of bike rentals. In this part of the Lab, you will look at a new data set and do forward and backward selection to look at feature selection.

Data Preprocessing Notes

Before you start, note that horsepower is missing for 5 cars in this dataset, and is coded as "?" in the file. To load the data, use the following code:

```
pd.read_csv("Auto.csv", na_values="?").dropna().reset_index(drop=True)
```

This reads the ? values as NA and then drops them as we typically do. After doing this, you should have 392 cars for both Parts 1 and 2.

In this analysis, we will treat cylinders and origin as categorical variables, and the rest of the variables as numeric. In this dataset, origin indicates if the car is from the US, Europe, or Japan. Number of cylinders we will explore in part 1a before encoding.

1. **[4 pts]** Data pre-processing + exploration:
 - a. As noted above, we are going to treat cylinders as a categorical variable. Count how many unique records there are for each cylinder type. Then, calculate and use a boxplot to show the the distribution of mpg for each cylinder count.
 - b. Create a new categorical variable called “cyl_rollup” and create the following categories
 - i. < 6 cylinders
 - ii. 6 cylinders
 - iii. 8 cylinders
 - c. Add this new category to the auto data frame and create the same plot as in part a, replacing cylinders with cyl_rollup. What do you observe?
 - d. Why should we exclude “name” from the dataset?
2. **[6 pts]** Now we will do forward and backward feature selection. For this section, use only the numeric variables in the data frame: ["displacement", "horsepower", "weight", "acceleration", "year"]. Use StandardScaler to scale the variables as we did in Lab4 and in class examples. Note that you will also need to create a KFold cross validation object for this section. Use the same KFold object for both parts a and b
 - a. Run forward selection with SequentialFeatureSelector, LinearRegression(), direction="forward", and n_features_to_select=k for k = 1, 2, ..., 4. For each value of k, print out which features are selected. This will tell you the order in which the features are selected.
 - b. Run backward selection with n_features_to_select=3. Do forward and backward selection choose the same three features? If they do, does that guarantee that they found the best model? If they don't, why not?
 - c. Look at the correlations between displacement, horsepower, and weight and calculate their VIFs, as we did in Lab 3. (Feel free to use statsmodels for this.) For variables with high VIF numbers, how can this affect the features chosen for Backward Selection? What about forward selection?

Part 2: Cross-Validation Approaches (10 pts)

In this section, we will use train/test splits and cross validation to test a couple of different models. For this section, please use the auto dataset that we used in Part 1, and use the following features (x variables) in your models. The y variable is always mpg:

- Model A: Numeric: [displacement, weight, year]
- Model B: Numeric: [displacement, weight, year], Categorical: [cyl_rollup, origin]

For each of these, use a pre-processing step that uses a standard scaler for each numeric variable and a one hot encoder for each categorical variable. (Use drop = “first”)

1. **[3pts]** First, we will build each of these models using a single train/test split, with 80% training and 20% testing.
 - a. Build Models A and B with LinearRegression using only the training data. Calculate the RMSE for both the training (in-sample) and testing (out of sample) data. Create a table that compares the train and test error for each of these Models.
 - b. Which model would you say is better and why?
2. **[3pts]** Now we will use k-fold cross validation with LinearRegression to build Models A and B
 - a. Build each model with 5-fold and 10-fold cross-validation.
 - b. For each model, compare the out of sample RMSE that was calculated in Question 1 (single train/test split), 5-fold cross-validation, and 10-fold cross validation. What patterns do you see?
 - c. Which model would you now say is better and why?
3. **[4pts]** Finally, we want to repeat questions 1 and 2 with 100 different random seeds, similar to what we did in the `feat_selection_pt3_cross_validation.ipynb` notebook. The code to do this is in Part 1 of this in class notebook. I would like you to generate the plot entitled *RMSE estimates across 100 random seeds* for Models A and B. Note: You do NOT have to do repeated k-fold or LOOCV for this exercise. You will need to make a slight change to the plotting code to do this.
 - a. Based on these plots, which model would you choose and why?
 - b. Is adding the categorical variables worth the extra complexity? (Note: There are multiple correct answers for this one!)

Part 3: Polynomial and Spline Regression (15 pts)

On 10/7, we predicted bike rentals from temperature only using polynomials and splines. We weren't able to get fully through the notebook. Here, we will pick up where we left off, and compare a more limited subset of models. A lot of the code in that notebook can be reused for this exercise.

Guidance for this section:

- For polynomial models, use `make_pipeline(StandardScaler(), PolynomialFeatures(d, include_bias=False), LinearRegression())`. The `StandardScaler` is not in the 10/7 notebook, but is helpful here for high degree polynomials.
 - For cubic splines, use `make_pipeline(SplineTransformer(degree=3, n_knots=k, knots="quantile"), LinearRegression())`
1. **[5 pts]** Using the bike sharing data, fit three models predicting daily bike rentals as a function of temperature. Use a single 80/20 train/test split.
 - a. Model A: Standard Linear Regression
 - b. Model B: Polynomial, degree = 9
 - c. Model C: Cubic Spline, knots = 7
 - d. Create a plot of temperature and bike rentals with fitted curves from each of these three models included. (Again, similar to our in class notebook) What do you observe?
 2. **[6pts]** Calculate the RMSE on in-sample (training) and out-of-sample (testing) data.
 - a. Which model has the smallest training RMSE and which has the smallest test RMSE?
 - b. Does decreasing the training error imply decreasing the test error? Use results from these models to support your answer.
 - c. Is there any evidence of overfitting or underfitting in any of these models?
 3. **[4 pts]** Rebuild Models A, B, and C, but this time use 5-fold cross-validation. Create a table comparing the following for each model: Training RMSE, Testing RMSE, k-fold RMSE. Does this give you more confidence in your answers around model complexity? Why or why not?

Part 4: The Bias-Variance Tradeoff (10 pts)

Now we are going to look more explicitly at the bias-variance tradeoff. As we discussed in class, increasing flexibility often reduces bias because the model can represent more complex relationships, but flexibility can increase variance because the fitted model is more sensitive to the particular training sample.

One thing we can ask after we build a model is: How would its predictions change if we had a different training sample? If we have a different sample of data, re-train our model, and there is a large difference in predictions at a fixed value of x_0 , this suggests that there is a lot of variance in the model. (Slides 28 and 29 of our Week 5/6 deck frame this.)

Bootstrapping is a way to explore this question. The bootstrapping approach you will want to use is as follows:

- Use the 80/20 train/test split of the data from Part 3. (We are using this for consistency)
- Use $B=500$
- Use a specified seed for your random number generator
- For each bootstrap simulation:
 - Sample with replacement from the training data
 - Refit your model with this new sample
 - Predict the value of y for $x=0.60$. (We want to predict every model on the same temperature, and we are choosing 0.60)
 - Save this predicted value for each of the simulations

1. **[6 pts]** Let's look at variance on the bike data for each of these models. Use the same train/test split you used in Part 3 to make the results directly comparable. For each of the three models in the previous question (Linear, Polynomial degree=9, and Cubic with 7 knots), do the bootstrap simulation described above.
 - a. Create a box plot of Predictions with each Model on the x-axis. Compare the distributions of predictions for each of the models. Which one is most stable? Which one appears most variable?
 - b. Compare prediction variability across the linear (A), polynomial (B), and spline (C) models. What does this tell you about the relationship between model flexibility and sensitivity to the training sample?
 - c. More generally speaking, suppose a model fit the training observations very closely, but the predictions at $\text{temp}=0.60$ change substantially across bootstrap samples. What does this tell you about the model's variance and its potential to generalize.
2. **[4 pts]** Putting it all together.
 - a. Look at the bootstrap results, as well as the train/test RMSE and cv results, and the fitted curves, describe the bias variance tradeoff across the linear (A), polynomial (B), and spline (C) models.
 - b. A teammate says: "We should just choose the model with the smallest training RMSE." Explain why this reasoning is incomplete. Use the ideas of bias, variance, overfitting, and generalization.

Deliverables

For each part, please include the following:

- All code used to complete the activities. This can be turned in as a single Jupyter notebook, multiple Jupyter notebooks, or one or more .py files. Up to you!
- For open ended questions, please include a few sentences or a paragraph answering the question. This text can be included in your Jupyter notebook as a markdown cell or a separate document.
- Per our policy on Generative AI detailed in the syllabus, if you use it to generate code, you need to flag how and where you used it, and how you ascertained the correctness of what you did. You are not allowed to use Generative AI for any of the written explanations or reflection questions. If you did not use Generative AI for any part of the assignment, please state that explicitly.